

Yash Saxena

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EDUCATION

Indraprastha Institute of Information Technology, Delhi

B.Tech in Computer Science and Applied Mathematics (CGPA: 8.60)

New Delhi, India

Aug 2024 – May 2028

City Montessori School

ISC Class XII – 97.75%

ICSE Class X – 97.2%

Lucknow, India

2024

2022

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, SQL

Technologies & Tools: Git, GitHub, Linux, Java Swing, JUnit, Maven, VSCode, MySQL Workbench, LaTeX, Figma

Core Competencies: Data Structures & Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems, Computer Organization, Human Computer Interaction, Automata Theory

WORK EXPERIENCE

Undergraduate Researcher – IIIT Delhi

May 2026 – Present

Guide: Dr. Debika Banerjee

New Delhi, India

- Investigated problems in **partition theory** by integrating theoretical mathematical foundations with computational experimentation and analysis. Engineered **Python-based recursive backtracking algorithms** to generate, explore, and evaluate complex combinatorial structures efficiently. Developed **interactive web visualizations** using **JavaScript** and **Chart.js** to dynamically represent and analyze **q-series truncation behavior** in real time.

PROJECTS

Skillr – Freelancing Platform (DBMS Project) | Database Design, MySQL, JS

Jan 2026 – Apr 2026

- Designed a **freelancing platform database** supporting users, projects, bids, contracts, and payment workflows.
- Built **ER model and normalized relational schema** with integrity constraints and indexing for efficient querying.
- Implemented **various SQL queries** including joins, aggregations, and nested queries to support core features.
- Developed **triggers and transaction workflows** to enforce business rules and handle concurrent operations safely.

University ERP System | Java, Java Swing, MySQL, JDBC, Maven, bcrypt | [GitHub](#)

Sep 2025 – Dec 2025

- Designed and developed a desktop-based **University ERP system** supporting three role-based user flows - **Students**, **Instructors**, and **Admins** - with end-to-end coverage of academic, administrative, and grading workflows.
- Architected a dual-database system separating authentication data **auth_db** from core application data **erp_db** to improve security, modularity, and maintainability.
- Designed a **modular, multi-tier system** architecture comprising **40+ backend services** and **27+ interactive GUI modules**, implementing secure authentication, role-based access control, transaction management, and backup/restore mechanisms to ensure scalability, maintainability, and fault tolerance.

Operating System Implementation | C, Linux, System Programming | [GitHub](#)

Aug 2025 – Dec 2025

- Implemented core operating system components in **C**, including a **program loader**, **interactive shell**, **process scheduler**, and **multithreading mechanisms**.
- Implemented low-level systems concepts such as **ELF file parsing**, **lazy loading via page faults**, **signals and pipes**, and multiple scheduling algorithms (**MLFQ**, **RR**).
- Implemented shell-commands like **wc**, **grep** and **MLFQ Scheduler** in **EGOS-2000** Environment

RISC-V Assembler and Simulator | Python, RISC-V ISA, Assembly Language | [GitHub](#)

Jan 2025 – Apr 2025

- Built a custom assembler and simulator for the **RV32I (32-bit RISC-V integer)** instruction set.
- Implemented an **assembler pipeline** that translates assembly to binary with **robust syntax validation**, **label resolution**, and **error reporting**.
- Developed a cycle-accurate simulator to execute binaries and generate detailed execution traces, supporting **15+ R-, L-, I-, S-, B-, and J-type** instructions.

LEADERSHIP & ACHIEVEMENTS

Batch Representative | Student Senate IIITD

Jan 2026 – Present

- Facilitated communication between students and administration to address academic issues and improve student experience.

Branch Representative | Student Council IIITD

Aug 2024 – May 2025

- Acted as a liaison between branch students and institutional authorities to resolve non-academic concerns.